

Mohan Zhang

mohanz@cs.unc.edu • [Homepage](#) • [Google Scholar](#) • [GitHub](#)

RESEARCH INTEREST

Efficient and Reliable ML

- Building intelligent agent systems for diverse applications (web navigation, robotic automation, *etc.*)
- Improving LLM scaling efficiency (via MoE, sparsification, *etc.*) and reliability.

EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

Starting From Aug 2025

- Ph.D. in Computer Science
- Advisor: [Prof. Tianlong Chen](#)

University of Science and Technology of China, Anhui, China

Sep 2021 – Jun 2025

- B.S. in Computer Science & Technology (GPA: 3.85/4.3)

PUBLICATIONS

([†]: Equal contribution)

CONFERENCES

Advancing MoE Efficiency: A Collaboration-Constrained Routing (C2R) Strategy for Better Expert Parallelism Design

Mohan Zhang[†], Pingzhi Li[†], Jie Peng, Mufan Qiu, Tianlong Chen

NAACL 2025, SAC award for Low-resource Methods for NLP

One Token Embedding Is Enough to Deadlock Your Large Reasoning Model

Mohan Zhang[†], Yihua Zhang[†], Jinghan Jia, Zhangyang Wang, Sijia Liu, Tianlong Chen

NeurIPS 2025

Bridging Language and Physics: Automated Design of Continuum Robots with Large Language Models

Jingyi Chen[†], **Mohan Zhang**[†], Laura Yao[†], Yingtai Ni, Jianmin Ji, Jie Peng, Song Wang, Tianlong Chen

Robotics: Science and Systems (RSS) 2026

Symbiotic Cooperation for Web Agents: Harnessing Complementary Strengths of Large and Small LLMs

Ruichen Zhang[†], Mufan Qiu[†], Zhen Tan[†], **Mohan Zhang**, Xiaopeng Lu, Jie Peng, Kaidi Xu, Leandro Z. Agudelo, Peter Zhenghao Qian, Tianlong Chen

CPAL 2026

Modalities Contribute Unequally: Enhancing Medical Multi-modal Learning through Adaptive Modality Token Re-balancing

Jie Peng[†], Jenna L. Ballard[†], **Mohan Zhang**, Sukwon Yun, Jiayi Xin, Qi Long, Yanyong Zhang, Tianlong Chen

ICML 2025

IN SUBMISSION/ PREPARING

([†]: Equal contribution)

JOURNALS

Precision Agriculture Autonomy with an Agent and Human Robot Coordination

Mohan Zhang[†], Xingjian Li[†], Yuhang Chen[†], Weilong He[†], Zhen Tan[†], Zhenghua Zhang[†], Jinhao Duan, Chris Reberg-Horton, Charles Fleming, Ron Alterovitz, Lirong Xiang, Tianlong Chen

► In this work, we introduce an LLM agent that enables transferable, multi-robot autonomy in agriculture by converting natural language commands into executable plans and robustly deploying them across diverse real-world conditions, paving the way for human-AI-robot collaboration in precision farming.

Submitted to Nature Communications

CONFERENCES

Measuring Real-World Prompt Injection Attacks in LLM-based Resume Screening

Mohan Zhang, Yuqi Jia, Zhen Tan, Steven Jiang, Neil Zhenqiang Gong, Tianlong Chen, Dawn Song

Under review, USENIX Security 2026

Repudiation Prompt Injection Attack to LLM-based Resume Screening

Zhen Xu, **Mohan Zhang**, Zhen Tan, Neil Zhenqiang Gong, Tianlong Chen

Submitted to NeurIPS 2026

TRUST: A Crowdsourcing Framework for Auditing Large Language Model Reasoning

Morris Yu-Chao Huang, Zhen Tan, **Mohan Zhang**, Pingzhi Li, Zhuo Zhang, Tianlong Chen

Submitted to COLM 2026

DOGe: Defensive Output Generation for LLM Protection Against Knowledge Distillation
Pingzhi Li, Zhen Tan, **Mohan Zhang**, Huaizhi Qu, Huan Liu, Tianlong Chen
arXiv:2505.19504

RESEARCH EXPERIENCES	The University of North Carolina at Chapel Hill , Chapel Hill, NC Remote Research Intern, mentored by Prof. Tianlong Chen	Feb 2024 – Jun 2025
AWARDS & SCHOLARSHIPS	Scholarships - SAC award for Low-resource Methods for NLP at NAACL 2025. - Outstanding Student Scholarship of USTC	2025 2022, 2023, 2024